Independent reviewer checklist

OHMETRY 2 MW DATA CENTRE ELECTRICAL ENGINEERING PROJECT. Package revision A6.
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FOR INDEPENDENT ENGINEERING REVIEW. THE PACKAGE IS PRELIMINARY. IT IS NOT CONSTRUCTION-READY AND NOT VERIFIED. NOTHING IN IT HAS BEEN APPROVED.

Mark each item PASS, FAIL, N/A or QUERY and record findings in REVIEW_FINDINGS_LOG.xlsx with the item reference. The checklist does not limit the review.

A. Scope, independence and conduct of the review

Ref	Check	Where	Result	Notes
R01	Confirm you are independent of the authors and have no stake in the package.	Reviewer declaration		
R02	Confirm the scope: educational and preliminary reference package, not a project design. Note anything claimed beyond that.	CUSTOMER_README sections 1, 13 to 16		
R03	Confirm nothing is labelled approved, verified or construction-ready.	All documents; run the search listed in REVIEW_GUIDE		
R04	Reproduce the package from the review package: install, run the test suite, regenerate the deliverables (REVIEW_GUIDE, REPRODUCE).	REVIEW_GUIDE		

B. Design basis

Ref	Check	Where	Result	Notes
R05	Check each design-basis parameter, unit and source; confirm the data status is honest (preliminary vs placeholder vs reference).	01_MASTER_DESIGN_BASIS.xlsx, config/project_config.yaml		
R06	Check the voltage and frequency basis (400/230 V,	ER-001, lv_voltage_frequency check		

	60 Hz is flagged as unusual); decide whether it is acceptable for the reference project.			
R07	Check the redundancy concept (2N paths, 2N UPS, N+1 generators) is internally consistent and what 'credible contingency' means here.	03_POWER_ARCHITECTURE, config redundancy section		
R08	Check the loading policy (transformer 90/100 %, UPS 80/100 %, generator 80/100 %) is applied to the derated rating and no overload is used.	docs/ calculation_methodology/ power_system_sizing.md		
R09	Check the design-target exception EX-001 is recorded correctly and never converts a WARNING to PASS.	config/ design_exceptions.yaml, validation/exceptions.py		

C. Load estimate and PUE

Ref	Check	Where	Result	Notes
R10	Check P and Q are added separately and S = sqrt(P^2+Q^2); check the UPS loss formula and distribution loss treatment.	docs/.../ load_estimation.md, calculations/load.py, 02_LOAD_ESTIMATION		
R11	Check the load schedule items, demand and power factors are reasonable for the reference project.	config/ load_schedule.yaml		
R12	Confirm the PUE is a design-point power ratio and not described as annual anywhere.	02, 18, training module 02		
R13	Check the A/B allocation and the path-out scenarios.	AB_ALLOCATION sheet, calculations/load.py		

D. Equipment sizing

Ref	Check	Where	Result	Notes
R14	Transformer: loading against the derated rating, contingency case, required rating for the target; confirm 3,150 kVA result.	04_TRANSFORMER_SIZING, calculations/sizing.py		
R15	Generators: PRP basis stated as assumed; kW and kVA both checked; N+1 and the double-contingency treatment; step loads reported NOT VERIFIED.	05_GENERATOR_SIZING, ER-010, ER-012		
R16	UPS: kW and kVA both checked; 2N logic; kVA governs at 0.95 power factor.	06_UPS_SIZING, ER-009		
R17	Battery: energy only; confirm no battery selection is implied.	07_BATTERY_SIZING		
R18	Derating: 1.00 placeholders are shown as placeholders everywhere and their effect is stated.	ER-011, derating_factors check		
R19	Check the size lists and suggested sizes are labelled illustrative.	config size lists		

E. Distribution

Ref	Check	Where	Result	Notes
R20	Design current bases for each feeder; busbar and busduct required current (about 4,136 A on path-out).	08, 11, 12, 13; calculations/distribution.py		
R21	Conductor resistance constants and the three-phase voltage drop formula; reactance placeholders and	docs/.../distribution.md, 09_VOLTAGE_DROP		

	their effect.			
R22	Confirm tabulated ratings, busduct and switchgear ratings are never assumed and read NOT VERIFIED.	08, 11, 12, 13		
R23	Check the cable selections are labelled illustrative and path B is assumed identical.	config/ distribution.yaml		

F. Short-circuit, protection and earthing

Ref	Check	Where	Result	Notes
R24	Check the IEC 60909 equations (utility, K_T, K_G, peak factor, thermal factor) against the standard; check the transformer-limited bound.	calculations/ short_circuit.py, docs/.../short_circuit_and_earthing.md		
R25	Confirm no utility, generator or site value is invented; cases that lack data read NOT CALCULATED.	10_SHORT_CIRCUIT, ER-019, ER-024		
R26	Check the adiabatic cable withstand constant and its flagged status.	08_CABLE_SIZING		
R27	Check the IEEE 80 grid resistance and tolerable voltage equations against the standard; confirm the screening wording.	calculations/ earthing.py, 15_EARTHING_BONDING		
R28	Confirm protection coordination, arc-flash and lightning compliance are never claimed.	14, 16 workbooks		

G. Status model and data override

Ref	Check	Where	Result	Notes
R29	Confirm a calculation on unverified inputs can never reach overall PASS; confirm a	validation/ verification.py, docs/.../status_model.md		

	preliminary WARNING or FAIL is kept.			
R30	Confirm every engineering check depends on the right inputs (the CHECK_INPUTS map).	validation/ verification.py		
R31	Import: confirm missing source, missing datasheet revision, unknown parameter, unsupported unit and implausible value reject the whole file; confirm unit conversion.	reporting/ data_import.py		
R32	Confirm the master design basis is never edited by an import and the history is append-only.	config/loader.py, manufacturer_data _overrides.yaml		
R33	Confirm MANUFACTURER _VERIFIED is reachable only through an import plus a recorded reviewer.	models/ data_status.py, cli record-verification		

H. Workbooks

Ref	Check	Where	Result	Notes
R34	Audit formulas in each workbook: no hard-coded critical values, correct references, correct units.	workbooks/ (all 20)		
R35	Enter a datasheet value without a source: confirm it is rejected. Enter with a source: confirm the calculation updates and the status becomes SUPPLIED. Add a reviewer: confirm VERIFIED.	INPUTS sheets		
R36	Confirm the overall status can only be PASS when the calculation passes and all inputs are verified (try it).	RESULTS sheets		

R37	Open in Microsoft Excel (the tests recalculated with LibreOffice only; Excel was not available to the authors, so this has never been done) and confirm results agree. Read WORKBOOK_LINT.md for the static lint.	All workbooks, WORKBOOK_LINT.md		
R38	Check the five input-register workbooks (14, 16, 17, 18, 19) do not produce a result without data and do not claim compliance.	14, 16, 17, 18, 19		

I. Python engine

Ref	Check	Where	Result	Notes
R39	Review the code against the methodology documents; check units, boundary conditions and error handling.	engine/src/ ohmetry_dc		
R40	Run the full test suite and confirm all tests pass; read the tests for the engineering-critical formulas (hand calculations).	engine/tests, QA_RESULTS.md		
R41	Check for hidden defaults: where an input is missing the engine must report it, not substitute a value.	calculations/*.py		
R42	Check configuration validation: required inputs, stage gating, value_ref resolution.	config/loader.py, models/		

J. Reports, drawings and registers

Ref	Check	Where	Result	Notes
R43	Read the load estimation report; confirm	reports/		

	statements, limitations and the agreement with the workbook.			
R44	Review the single-line diagram: architecture, tags, no unverified value shown as fact; decide path independence and transfer scheme.	drawings/		
R45	Check the assumptions register, standards register (all NOT_VERIFIED) and manufacturer-data register for completeness.	07_registers		
R46	Verify the outstanding items list is complete; add anything missing.	OUTSTANDING_VERIFICATION_ITEMS		

K. Specifications and training

Ref	Check	Where	Result	Notes
R47	Review each specification template: standards named correctly, requirements sound, test lists complete, placeholders clear, no invented numbers.	08_specifications		
R48	Review the training modules, worked examples, exercise answers and question bank for technical accuracy and fair wording.	09_training		
R49	Confirm the certificate states it is not accredited.	Completion_Certificate_Template		

L. Documentation and commercial

Ref	Check	Where	Result	Notes
R50	Check the customer README is accurate against	CUSTOMER_README		

	the package and states limitations honestly.			
R51	Note that the licence draft is outside this review: it needs a legal professional.	LICENCE_COMMERCIAL		
R52	Check the master index and manifest: every file has an id, revision and hash; cross-references are right.	MASTER_INDEX, MANIFEST.json		

M. Revision A6 corrective actions (read CORRECTIVE_ACTION_REPORT first)

Ref	Check	Where	Result	Notes
R53	Architecture: confirm each concept symbol on SLD-001 (generator breakers, ACB-TX / ACB-GEN interlock, PMS, static and maintenance bypass, battery isolator, boundaries) is labelled concept and not designed; confirm independence of utility A and B is claimed nowhere; confirm the IT load is one dual-corded load.	drawings/, models/architecture.py, 03_POWER_ARCHITECTURE sheet FUNCTIONS, ER-028		
R54	Distribution: confirm no derating factor of 1.00 reads as verified, required ratings are labelled minimums, and installation method, parallel arrangement and current sharing are reported as missing; confirm a grouping factor of 1.00 for parallel runs is called out.	check cable_derating_basis, 08_CABLE_SIZING, docs/.../distribution.md, ER-029		
R55	Short-circuit: confirm the transformer-limited	checks short_circuit_levels,		

	figure is never described as the bus fault level and never used to select a rating; confirm cases without data read NOT CALCULATED and arc-flash is out of scope.	arc_flash_scope, ER-024, ER-030		
R56	Earthing and lightning: confirm earthing is labelled preliminary screening only (not a mesh and step study) and that no IEC 62305 compliance is claimed.	checks earthing_framework, lightning_scope, ER-031		
R57	Verification: confirm a reviewer name alone never marks a value verified (name, organisation, date, evidence reference and scope are required), the log is append-only, and the text says the name is not authenticated.	models/ attestation.py, verification_log.yaml, ER-032		
R58	Standards and specifications: confirm every standard named in a specification is in the register, no clause number is cited and editions are blank; review the vendor data schedule, compliance schedule, FAT, SAT, commissioning and integrated systems test sections.	07_registers/ standards_register.yaml, 08_specifications, ER-033		
R59	Document control: confirm revision A6 everywhere, no generated file from another revision, the ER log is reconciled (ER-028 to ER-034), the Excel limitation	MASTER_INDEX, ENGINEERING_REVIEW_LOG, WORKBOOK_LINT.md, ER-034		

	is stated, and WORKBOOK_LINT.md has been read.			
R60	Generator contingency (item 1): confirm the combined case (one path out and one engine out) is judged, the kW result reads WARNING (83.98 % of 3,200 kW) and is not converted to PASS, and that the contingency basis is flagged as the engineer's to select and document.	checks generator_loading_kw, generator_contingency_basis, ER-035		
R61	Transformer (item 2): confirm EX-001 is reconciled with the exact 90.97 % and its 0.03-point margin, the 90 % target and 100 % limit are unchanged, and that 3,150 and 4,000 kVA are shown side by side with nothing selected (note the 4,000 kVA fault-level consequence).	check transformer_loading, equipment_options, workbook 04 OPTIONS, ER-036		
R62	UPS (item 3): confirm kW (80.0 %, PASS) and kVA (84.2 %, WARNING) are separate and the illustrative 3,000 kVA option is shown.	checks ups_loading_kw, ups_loading_kva, workbook 06 OPTIONS		
R63	Load model (item 4): review the cooling redundancy, whole-unit A/B allocation, loss assumptions and the sensitivity table; confirm PUE is treated as an output and no input is adjusted to reach a target.	sizing summary, workbook 02 SENSITIVITY, check load_pue_vs_target, ER-037		
R64	Release gate:	CORRECTIVE_A		

	confirm the release state is DEVELOPMENT, nothing is marked approved, verified or construction-ready, and the corrective-action report agrees with the repository (re-run the tests and the key-number check).	CTION_REPORT, 11_qa/QA_RESULTS.md		

Sign-off

Field	Entry
Reviewer name and organisation	
Qualification and registration (if any)	
Independence declaration	I have no interest in the package or its sale and was not involved in producing it. [ ] Confirmed
Scope reviewed	
Date of review	
Overall conclusion (reviewer's own words)	
Signature	